1 Introduction

At the most basic level, a natural language model is a function that takes as input a sequence of words and attempts to predict the word that follows. This prediction takes the form of a probability distribution over the set of all possible words, indicating the level of confidence the model has that any given word might follow. This model generates text on a given input by sampling from the output distribution. Once a word is generated, it is concatenated to the end of the original input, leading to a new input for the model to once again sample from. Repeating this process yields output text as desired.

More precisely, text is segmented into *tokens*. Although tokens are often words, they take on many other forms: punctuation, prefixes and postfixes, single characters. Ideally, these tokens would allow for a model that is not restricted to one language or type of text, and could make meaningful predications on any form of structured text with sufficient training data (other languages, code, etc.). We will assume that the set $\mathcal{X}$ of tokens for our model is chosen beforehand, but note that determining this set is an important part of engineering an efficient language model.

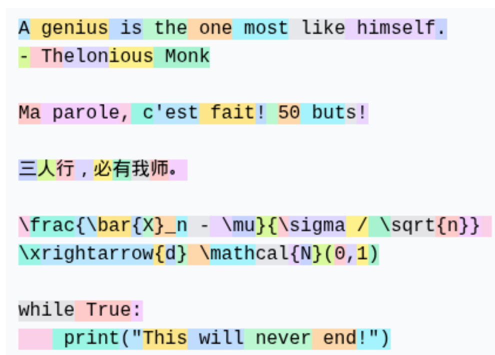

Figure 1: GPT-4 tokenizer on sample text.

Let n be the number of tokens in the sequence of text the model takes in as input, known as *context length*. The model can therefore be seen as a function $f : \mathcal{X}^n \to \Delta(\mathcal{X})$, where $\Delta(\mathcal{X})$ is the set of probability distributions over $\mathcal{X}$. To develop a model, we design the model's parametric form $f(\boldsymbol{X} \mid \theta)$ and use likelihood estimation on a large set of sample data to determine parameters θ for which model performs sufficiently well. These are not simple tasks: for any decent model, f is extremely complex and highly nonlinear, and the size of $\mathcal{X}$, n, and θ are several orders of magnitude beyond the regime on which classical methods in statistics are effective.

Model	Tokens $\mathcal{X}$	Context n	Parameters θ	Release
BERT (Large)	30000	512	340 Million	2018
GPT-2	50257	1024	1.5 Billion	2019
GPT-3	50257	2048	175 Billion	2020
Llama 3	128000	128000	405 Billion	2024

Table 1: Size of $\mathcal{X}$, n, and θ of some modern language models.

Assuming that there exists methods to optimize parameters θ with respect to model performance, this approach to developing a model is inherently powerful. Models can first be designed using intuition about how a machine may interpret and predict language under a mathematical framework. If the model performs sufficiently well after training, the goal is accomplished. From a purely predictive perspective, it is not necessary that the learned internal mechanisms resemble the original intuition motivating the architechture.

Earlier approaches to language modeling relied heavily on

Introduced in 2017, the transformer architecture resolves this issue through a different mechanism that allows every token to directly interact with every other token in the input.

2 The Transformer

This section provides a precise description of the architecture of the transformer model as outlined in (ATTENTION IS ALL YOU NEED). For each step, a brief heuristic justification of the design choice is given based on intuition about how a machine might use the architecture to model language effectively, however it is important to keep in mind that FINISH

Throughout, any *weight matrix* W or *bias vector* b is a learned parameter of the model, and any other value is either an input or an intermediary value. In a computational spirit, we will use the "$\leftarrow$" symbol to indicate variable assignment. Vector arrows will be used in order to clarify when individual rows or columns of matrices are in question.

2.1 The Softmax Function

2.2 Embedding and Positional Encoding

We assign each token in $\mathcal{X}$ is assigned a unique number from 1 to $|\mathcal{X}|$ and represent it as one-hot vector[1]. Thus, our input $X = \left(\vec{X}_1, ..., \vec{X}_n\right) \in \mathcal{X}^n$ is represented as a $|\mathcal{X}| \times n$ matrix, where each column is a one-hot encoding of a token.

Modeling and learning are much more difficult in high-dimensional discrete spaces, as the tools of calculus are not available. Hence, we choose an *embedding dimension* d which allows for a continuous representation of token "meaning" in $\mathbb{R}^d$. One interpretation motivating this embedding space is the idea that orthogonal directions could encode different concepts or ideas, and that linear combinations of vectors along these directions could represent the meaning of a given token.

[1] A one-hot encoding is a vector that has a 1 in the position corresponding to the number it represents, and 0 everywhere else.

A matrix $W_E \in \mathbb{R}^{d\times|\mathcal{X}|}$ contains all token embeddings, where column i is the embedding for token i. For our input X, the columns of the matrix multiplication $W_E X$ give the corresponding embeddings for each token. We also allow for the model to obtain information of the position of a token in the sequence. This is given by the matrix W_P, where column i contains the encoding for position i. This is added to the token embeddings to obtain

$$E \leftarrow W_E X + W_P,$$

which contains all token-level information that is independent of any interactions between tokens.

2.3 The Attention Layer

Given that words in any language rarely have a unique meaning independent of context, it is essential to allow tokens to "communicate" with each other. The attention layer modifies the positions of tokens in embedding space to better suit their meaning in context.

2.3.1 Single-Headed Attention

Let $W_K, W_Q \in \mathbb{R}^{d_h\times d}$, where d_h is known as the *head dimension*. Given the previously obtained token embeddings E, we compute

$$Q \leftarrow W_Q E, \qquad K \leftarrow W_K E,$$

called the *query matrix* and *key matrix*, respectively. Note that the columns of these matrices are transformations of embedding vectors: $\vec{Q}_i = W_Q \vec{E}_i$ and $\vec{K}_i = W_K \vec{E}_i$. The query $\vec{Q}_i$ represents what information the token is looking for from other tokens in order to refine its contextual meaning. The key $\vec{K}_i$ represents what kinds of information the token can provide to other tokens. Both these vectors are projections of their corresponding token embeddings into a shared lower-dimensional space $\mathbb{R}^{d_h}$, specifically used for relational matching.

To capture the extent to which the key $\vec{K}_j$ is relevant to query $\vec{Q}_i$, we compute the *attention score* $s_{ij} = \vec{Q}_i \cdot \vec{K}_j$, where larger values correspond to alignment in the query-key space. Thus, the matrix

$$S \leftarrow \frac{Q^T K}{\sqrt{d_h}}$$

has scaled attention score $s_{ij}/\sqrt{d_h}$ as entry (i, j). This scaling factor keeps dot-product similarity "dimension-invariant", meaning scores stay at a consistent scale regardless of head dimension.

Given a query, we would like these scores to correspond to weights indicating the relative importance of each key's token for updating the embedding of the queried token. We therefore apply the softmax function column-wise to S giving

$$A \leftarrow \mathrm{softmax}(S).$$

This completes the conversion of token embeddings into a normalized distribution over which tokens are most relevant to each other.

Returning to the original embeddings E, let $W_V \in \mathbb{R}^{d\times d}$. we compute the *value matrix*

$$V \leftarrow W_V E.$$

The columns $\vec{V}_i = W_V \vec{E}_i$ represent the information each token contributes when it is used by other tokens. Recalling that the columns $\vec{A}_i$ of the matrix A contain the attention weights assigned by token i, the update to a token's embedding is given by the weighted sum of the value vectors according to weights $\vec{A}_i$. In matrix form, this can be written as

$$\Delta E \leftarrow VA.$$

This update is then applied to E to obtain the final output of the attention layer,

$$E_{\text{out}} \leftarrow E + \Delta E.$$

2.3.2 Multi-Headed Attention

We wish to use multiple single-headed attention layers in parallel to allow different heads to specialize in different types of relationships and patterns within the input sequence. Let h be the number of heads in this multi-headed attention layer. We choose the head dimension d_h such that $d = hd_h$.

As before, for each head, we have matrices $W_Q^{(i)}, W_K^{(i)} \in \mathbb{R}^{d_h \times d}$ and compute

$$A^{(i)} \leftarrow \operatorname{softmax}\left(\frac{Q^T K}{\sqrt{d_h}}\right).$$

Allowing for each $W_V^{(i)}$ matrix to have size $d \times d$ would result in hd^2 parameters across all heads. In practice, we would like the parameter count to be comparable to W_Q and W_K, so we force W_V to have the low-rank structure

$$W_V^{(i)} = W_{V_l}^{(i)} W_{V_r}^{(i)},$$

where $W_{V_l}^{(i)} \in \mathbb{R}^{d \times d_h}$ and $W_{V_r}^{(i)} \in \mathbb{R}^{d_h \times d}$. Across all heads, this contributes $2dd_h$ parameters.

While this factored matrix could be applied per-head, each attention layer instead computes $W_V^{(i)} A$, which are then concatenated vertically and multiplied by the horizontal concatenation of $W_{V_l}^{(i)}$ matrices, giving[2]

$$\Delta E \leftarrow \underbrace{\begin{bmatrix} W_{V_l}^{(1)} & \cdots & W_{V_l}^{(h)} \end{bmatrix}}_{W_O} \begin{bmatrix} W_{V_r}^{(1)} A \\ \vdots \\ W_{V_r}^{(h)} A \end{bmatrix},$$

which is equivalent to computing $\Delta E^{(i)} \leftarrow W_V^{(i)} A$ for every head and summing the results. Because of this, we will consider the horizontal concatenation W_O to be a separate matrix that is applied after concatenating the results from each individual head. For each head, we will consider the $W_{V_r}^{(i)}$ matrix to simply be the $W_V^{(i)}$ matrix applied in the single-headed case.

Adding this to the original embeddings gives the final output for the multi-headed attention,

[2]This allows the transformation to be expressed as a composition of two learned linear maps, which is computationally more efficient and better structured for parameter optimization than a product of two matrices.

$$E_{\text{out}} \leftarrow E + \Delta E.$$

2.3.3 Masking and LayerNorm

MASKING LAYERNORM

2.4 Feed Forward Layer

Once a token has been contextualized using attention, it carries information from across the sequence, but this information is still only combined linearly through a weighted sum. The feed forward layer allows for each token to independently introduce nonlinear transformations in its embedding, converting this context into more expressive and higher-level representations. This is done using a *multi-layer perceptron*.

Let E be input embeddings that have been processed by an attention layer. The multi-layer perceptron aims to model how a biological brain might process information using neurons. We choose a feed-forward dimension d_f, which we will interpret to be the number of neurons in this layer.

Each neuron processes information using an affine transformation of the input data. This is then fed into a non-linear *activation function* σ which aims to model how a neuron might fire. We use $\sigma(x) = \max(0, x)$, known as ReLU, although there are many other options. The *activation* of neuron i for some embedding $\vec{E}_j$ is the scalar value given by

$$n_{ij} \leftarrow \sigma\left(\overrightarrow{W}_{\text{in}}^{(i)} \cdot \vec{E}_j + b_{\text{in}}^{(i)}\right).$$

Letting W_{in} to be the matrix with weights $\overrightarrow{W}_{\text{in}}^{(j)}$ as columns and b_{in} to be the column vector with components $b_{\text{in}}^{(i)}$, this can be expressed in matrix form as

$$N \leftarrow \sigma\left(W_{\text{in}} E + b_{\text{in}} \mathbf{1}_{d_f}^T\right),$$

where σ is applied component-wise, and $\mathbf{1}_{d_f} \in \mathbb{R}^{d_f}$ is the column vector of all ones.

Similar to attention, an update to each embedding is calculated using an affine transformation of the embeding's corresponding neuron activations. In parallel, this gives

$$\Delta E \leftarrow W_{\text{out}} N + b_{\text{out}},$$

which is added to the input to give

$$E_{\text{out}} = E + \Delta E.$$

2.5 Unembedding

In practice, embeddings are fed through many iterations of transformer blocks (a multi-headed attention layer followed by a feed forward layer), each with separate parameters. After this processing, we wish to convert each embedding $\vec{E}_i$ into a probability distribution over tokens $\mathcal{X}$, indicating the level of confidence that each token in the vocabulary could be the next token at position i.

This is done by applying the unembedding matrix $W_U \in \mathbb{R}^{|\mathcal{X}| \times d}$ to $\vec{E}_i$, followed by a softmax. In matrix form,

$$Y = \text{softmax}(W_U E + b_U \mathbf{1}^T),$$

where softmax is applied column-wise. Note that each column $\vec{Y}_i$ isthe prediction for the next token at each position i in the input sequence, ignoring tokens that come after position i. This is useful for model training purposes, but for generation purposes we only use the distribution $\vec{Y}_n$ corresponding to the last input token.

2.6 Summary

3 Attention Scaling

3.1 Insufficiency of Standard Softmax

Consider the standard softmax function applied to a sequence of score vectors $s^{(n)} \in \mathbb{R}^n$ with uniformly bounded components $m \leq s_j^{(n)} \leq M$ for $m, M \in \mathbb{R}$. For a fixed j,

$$\text{softmax}(s)_j = \frac{e^{s_j}}{\sum_{k=1}^n e^{s_k}} \leq \frac{e^M}{ne^m} = \frac{e^{M-m}}{n},$$

as well as

$$\text{softmax}(s)_j \geq \frac{e^{m-M}}{n}.$$

Then, we must have that $\text{softmax}(s)_j \to 0$ as $n \to \infty$.

more TODO

3.2 Scalable Softmax

4 Simplex-Like Geometry

4.1 Setup and Motivation

Summary of Chen et Al.

Consider a simplified version of unmasked single-headed attention in which the head dimension equals the embedding dimension and $W_K = W_Q = W_V = I_d$. For a collection of token embeddings $x_1, ..., x_n \in \mathbb{R}^d$, define the normalized vectors $u_i = x_i / \|x_i\|$. The scaled attention scores for the normalized vectors u_i are thus given by

$$s_{ij} = \beta \langle u_i, u_j \rangle,$$

where $\langle \cdot, \cdot \rangle$ denotes the standard inner product on $\mathbb{R}^d$, and β is a scaling factor to be chosen. The corresponding attention matrix A has entries

$$a_{ij} = \frac{e^{s_{ij}}}{Z_i}, \qquad Z_i = \sum_{k=1}^n e^{s_{ik}}.$$

The main object at study is the nonlinear operator $\text{ATT} : \mathbb{R}^d \to \mathbb{R}^d$, defined as

$$\mathrm{ATT}(u_i) = \sum_{j=1}^{n} a_{ij} u_j.$$

Finally, recalling that in transformer architechtures, the attention layer computes an update to the original embeddings through a residual connection, we define the updated embeddings and their normalized versions

$$x_i' = \mathrm{ATT}(u_i) + \alpha x_i, \qquad u_i' = \frac{x_i'}{\|x_i'\|},$$

where $\alpha \geq 0$ is the strength of the residual connection. Geometrically, α interpolates between purely attention-driven dynamics and identity-preserving behaviour.

4.2 Critical Scaling

We are interested in how the attention scaling β affects the geometry of the token embeddings, namely how pairwise angles evolve under the attention update. Equivalently, this amounts to studying the relationship between $\langle u_i', u_j' \rangle$ and $\langle u_i, u_j \rangle$.

Consider a further simplified setup in which $\langle u_i, u_j \rangle = p \in (0, 1)$ if $i \neq j$ and $\|u_i\|^2 = q$. The key insight is that the behaviour of attention is governed by the normalization factor Z_i, which simplifies to

$$Z_i = e^{\beta} + (n-1)e^{p\beta}$$

under these assumptions. Thus, the asymptotic behaviour of Z_i depends on the competition between the self-attention term and the aggregate contribution of attention from all other tokens. Under the scaling $\beta = \gamma \log n$,

$$e^{\beta} = n^{\gamma}, \qquad (n-1)e^{p\beta} \asymp n^{1+p\gamma}.$$

Hence, the dominant contribution to Z_i depends on the comparison between the exponents γ and $1 + p\gamma$. The critical threshold occurs when $\gamma = \frac{1}{1-p}$. This reveals that there are three regimes under which attention operates. In the subcritical regime $\gamma < \frac{1}{1-p}$, attention weights become asymptotically uniform, so each token moves toward the global average. In the supercritical regime $\gamma > \frac{1}{1-p}$, the self-attention term dominates, and attention operator converges to the identity map. In the critical regime $\gamma = \frac{1}{1-p}$, both terms are balanced, leading to nontrivial dynamics as desired for an effective attention layer.

In, Theorem 4.2.1 we generalize this setup beyond the symmetric simplex setting by allowing pairwise inner products to vary within fixed bounds. We show that the attention operator acts asymptotically as a contraction in the subcritical regime and as an identity in the supercritial regime. Corollary 4.2.1 provides exact results for the limiting inner product for the simplex case. These results demonstrate that the scaling $\beta = \Theta(\log n)$ is intrinsic to the behaviour of attention mechanisms rather than an artifact of the simplex geometry.

Theorem 4.2.1 (Phase Transition in Attention Geometry): Assume there exist constants $q_1, q_2 > 0$ and $p_1, p_2 \in (0, 1)$ such that $q_1 \leq \|x_i\|^2 \leq q_2$ and $p_1 \leq \langle u_i, u_j \rangle \leq p_2$ for any $i \neq j$, with $p_1 = \langle u_i, u_j \rangle$ for some i, j. Let $\beta = \gamma \log n$ with $\gamma > 0$.

1. If $\gamma < \frac{1}{1-p_1}$ then there is a constant $\varepsilon > 0$ depending on α, p_2, q_1, q_2 such that

$$\liminf_{n\to\infty} \min_{i\neq j} \langle u_i', u_j' \rangle \geq p_1 + \varepsilon.$$

2. If $\gamma > \frac{1}{1-p_2}$, then for any i,

$$\lim_{n\to\infty} \langle u_i', u_j' \rangle = \langle u_i, u_j \rangle.$$

We will first prove a series of lemmas.

Lemma 4.2.1: For any $i \in \{1, ..., n\}$, we have

$$Z_i = \begin{cases} (1 + o_n(1)) \cdot \left(\sum_{k\neq i} e^{s_{ik}}\right) & \gamma < \frac{1}{1-p_1}, \\ (1 + o_n(1)) \cdot e^{\beta} & \gamma > \frac{1}{1-p_2}. \end{cases}$$

Proof. Recall $s_{ik} = \beta \langle u_i, u_k \rangle$ and $\beta = \gamma \log n$, giving

$$Z_i = \sum_{k=1}^{n} e^{s_{ik}} = n^{\gamma} + \sum_{k\neq i} n^{\gamma\langle u_i, u_j \rangle}$$

If $\gamma < \frac{1}{1-p_1}$, then $\gamma - \gamma p_1 - 1 < 0$. Using $\langle u_i, u_j \rangle \geq p_1$,

$$\frac{n^{\gamma}}{\sum_{k\neq i} n^{\gamma\langle u_i, u_j \rangle}} \leq \frac{n^{\gamma}}{(n-1) n^{\gamma p_1}} = n^{\gamma - \gamma p_1 - 1} \cdot \frac{1}{1 - 1/n}$$

which goes to 0 as $n \to \infty$. Hence,

$$Z_i = \left(1 + \frac{n^{\gamma}}{\sum_{k\neq i} n^{\gamma\langle u_i, u_j \rangle}}\right) \left(\sum_{k\neq i} n^{\gamma\langle u_i, u_j \rangle}\right) = (1 + o_n(1)) \left(\sum_{k\neq i} e^{s_{ik}}\right).$$

Similarly, if $\gamma > \frac{1}{1-p_2}$, then $\gamma p_2 + 1 - \gamma < 0$. Using $\langle u_i, u_j \rangle \leq p_2$,

$$\frac{\sum_{k\neq i} n^{\gamma\langle u_i, u_j \rangle}}{n^{\gamma}} \leq \frac{(n-1) n^{\gamma p_2}}{n^{\gamma}} = n^{\gamma p_2 + 1 - \gamma} (1 - 1/n)$$

which goes to 0 as $n \to \infty$. Hence,

$$Z_i = \left(1 + \frac{\sum_{k\neq i} n^{\gamma\langle u_i, u_j \rangle}}{n^{\gamma}}\right) (n^{\gamma}) = (1 + o_n(1)) \cdot e^{\beta}.$$

□

Lemma 4.2.2: If $\gamma > \frac{1}{1-p_2}$, then for any $i = \{1, ..., n\}$,

$$\mathrm{ATT}(u_i) = u_i + o_n(1),$$

where $\boldsymbol{o}_n(1)$ does not depend on i.

PROOF. By the previous lemma,

$$\mathrm{ATT}(u_i) = Z_i^{-1}\left(e^{\beta}u_i + \sum_{j\neq i} e^{s_{ij}}u_j\right) = (1 + o_n(1))\left(u_i + e^{-\beta}\sum_{j\neq i} e^{s_{ij}}u_j\right),$$

using the fact that $(1 + o_n(1))^{-1} = 1 + o_n(1)$. Since $\|u_j\| = 1$ for all j,

$$\left\|e^{-\beta}\sum_{j\neq i} e^{s_{ij}}u_j\right\| \leq e^{-\beta}\sum_{j\neq i} e^{s_{ij}} \leq n^{-\gamma}\cdot n^{\gamma p_2}(n-1)$$

which goes to 0 as $n \to \infty$, independent of i. □

Lemma 4.2.3: For any $i \in \{1, ..., n\}$,

1. If $\gamma < \frac{1}{1-p_1}$,

$$\|x_i'\|^2 \leq \alpha^2\|x_i\|^2 + 2\alpha\|x_i\|p_2 + p_2 + o_n(1).$$

2. If $\gamma > \frac{1}{1-p_2}$,

$$\|x_i'\|^2 = (\alpha\|x_i\| + 1)^2 + o_n(1).$$

In both cases, $o_n(1)$ does not depend on i.

PROOF. Using the definition of the update rule, expanding the inner product gives

$$\|x_i'\|^2 = \alpha^2\|x_i\|^2 + 2\alpha\|x_i\|\langle u_i, \mathrm{ATT}(u_i)\rangle + \|\mathrm{ATT}(u_i)\|^2.$$

For the case of $\gamma < \frac{1}{1-p_1}$, Lemma 3.1 gave $Z_i = (1 + o_n(1))\left(\sum_{j\neq i} e^{s_{ij}}\right)$. Hence,

$$\begin{aligned}\langle u_i, \mathrm{ATT}(u_i)\rangle &= \langle u_i, \frac{1}{Z_i}\sum_{j=1}^{n} e^{s_{ij}}u_j\rangle \\ &= \frac{1}{Z_i}\sum_{j=1}^{n} e^{s_{ij}}\langle u_i, u_j\rangle \\ &\leq \frac{1}{Z_i}\left(e^{\beta} + p_2\sum_{j\neq i} e^{s_{ij}}\right) \\ &= p_2 + o_n(1),\end{aligned}$$

since $e^{\beta}/\sum_{j\neq i} e^{s_{ij}} = o_n(1)$, as shown previously. Similarly,

$$
\begin{aligned}
\|\mathrm{ATT}(u_i)\|^2 &= \langle \frac{1}{Z_i}\sum_{k=1}^{n} e^{s_{ik}} u_k, \frac{1}{Z_i}\sum_{l=1}^{n} e^{s_{il}} u_l \rangle \\
&= \frac{1}{Z_i^2}\sum_{k=1}^{n}\sum_{l=1}^{n} e^{s_{ik}+s_{il}} \langle u_k, u_l \rangle \\
&\leq \frac{1}{Z_i^2}\left(e^{2\beta} + 2p_2 e^{\beta}\sum_{j\neq i} e^{s_{ij}} + p_2 \sum_{k\neq i}\sum_{l\neq i} e^{s_{ik}+s_{il}} \right) \\
&= p_2 + o_n(1).
\end{aligned}
$$

For the case of $\gamma > \frac{1}{1-p_2}$, applying Lemma 4.2.2 directly gives

$$
\begin{aligned}
\|x_i'\|^2 &= \alpha^2\|x_i\|^2 + 2\alpha\|x_i\|\langle u_i, u_i + \boldsymbol{o}_n(1)\rangle + \langle u_i + \boldsymbol{o}_n(1), u_i + \boldsymbol{o}_n(1)\rangle \\
&= \alpha^2\|x_i\|^2 + 2\alpha\|x_i\| + 1 + o_n(1) \\
&= (\alpha\|x_i\| + 1)^2 + o_n(1).
\end{aligned}
$$

□

Lemma 4.2.4: For $i, j \in \{1, ..., n\}$ with $i \neq j$,

1. If $\gamma < \frac{1}{1-p_1}$,

$$\langle x_i', x_j' \rangle \geq p_1(\alpha\|x_i\| + 1)(\alpha\|x_j\| + 1) + o_n(1).$$

2. If $\gamma > \frac{1}{1-p_2}$,

$$\langle x_i', x_j' \rangle = (\alpha\|x_i\| + 1)(\alpha\|x_j\| + 1)\langle u_i, u_j \rangle + o_n(1).$$

In both cases, $o_n(1)$ does not depend on i.

Proof. Expand the inner product to give

$$
\begin{aligned}
\langle x_i', x_j' \rangle = {} & \alpha^2\|x_i\|\|x_j\|\langle u_i, u_j\rangle + \alpha\|x_i\|\langle u_i, \mathrm{ATT}(u_j)\rangle \\
& + \alpha\|x_j\|\langle u_j, \mathrm{ATT}(u_i)\rangle + \langle \mathrm{ATT}(u_i), \mathrm{ATT}(u_j)\rangle.
\end{aligned}
$$

If $\gamma < \frac{1}{1-p_1}$, by following a similar procedure as in Lemma 4.2.3 we can instead bound the terms in question from below, i.e.

$$
\begin{aligned}
\langle u_i, \mathrm{ATT}(u_j)\rangle &= \langle u_i, \frac{1}{Z_i}\sum_{j=1}^{n} e^{s_{jk}} u_k \rangle \\
&\geq \frac{1}{Z_i}\left(p_1 e^{\beta} + p_1 \sum_{j\neq i} e^{s_{ij}} \right) \\
&= p_1 + o_n(1),
\end{aligned}
$$

and

$$
\begin{aligned}
\langle \mathrm{ATT}(u_i), \mathrm{ATT}(u_j)\rangle &= \langle \frac{1}{Z_i}\sum_{k=1}^{n} e^{s_{ik}}u_k, \frac{1}{Z_i}\sum_{l=1}^{n} e^{s_{jl}}u_l\rangle \\
&\geq \frac{1}{Z_iZ_j}\left(p_1e^{2\beta} + p_1e^{\beta}\sum_{k\neq i} e^{s_{ik}} + p_1e^{\beta}\sum_{l\neq j} e^{s_{jl}} + p_1\sum_{k\neq i}\sum_{l\neq i} e^{s_{ik}+s_{jl}}\right) \\
&= p_1 + o_n(1).
\end{aligned}
$$

Hence,

$$
\begin{aligned}
\langle x_i', x_j'\rangle &\geq \alpha^2\|x_i\|\|x_j\|p_1 + \alpha\big(\|x_i\| + \|x_j\|\big)p_1 + p_1 + o_n(1) \\
&= p_1(\alpha\|x_i\| + 1)\big(\alpha\|x_j\| + 1\big) + o_n(1).
\end{aligned}
$$

If $\gamma > \frac{1}{1-p_2}$, Lemma 4.2.2 can be applied to obtain

$$
\begin{aligned}
\langle x_i', x_j'\rangle &= \alpha^2\|x_i\|\|x_j\|\langle u_i, u_j\rangle + \alpha\|x_i\|\langle u_i, u_j + \boldsymbol{o}_n(1)\rangle \\
&\quad + \alpha\big\|x_j\big\|\langle u_j, u_i + \boldsymbol{o}_n(1)\rangle + \langle u_i + \boldsymbol{o}_n(1), u_j + \boldsymbol{o}_n(1)\rangle \\
&= (\alpha\|x_i\| + 1)\big(\alpha\|x_j\| + 1\big)\langle u_i, u_j\rangle + o_n(1).
\end{aligned}
$$

$\square$

We now prove Theorem 4.2.1.

Proof. First, consider the case where $\gamma < \frac{1}{1-p_1}$. We can further simplify the result in Lemma 4.2.3 by completing the square, giving

$$
\begin{aligned}
\|x_i'\|^2 &\leq \alpha^2\|x_i\|^2 + 2\alpha\|x_i\|p_2 + p_2 + o_n(1) \\
&= \alpha^2\|x_i\|^2 + 2\alpha\|x_i\| + 1 - 2\alpha\|x_i\| - 1 + 2\alpha\|x_i\|p_2 + p_2 + o_n(1) \\
&= (\alpha\|x_i\| + 1)^2 - (1-p_2)(2\alpha\|x_i\| + 1) + o_n(1) \\
&\leq (\alpha\|x_i\| + 1)^2 - (1-p_2)(2\alpha\sqrt{q_1} + 1) + o_n(1),
\end{aligned}
$$

where the last inequality follows from the fact that $\|x_i\| \geq \sqrt{q_1}$. Thus, there is some constant $\delta > 0$ depending on α, p_2, q_1, q_2 such that

$$
\frac{1}{\|x_i\|} \geq \frac{1+\delta}{\alpha\|x_i\| + 1} + o_n(1).
$$

An equivalent formulation of the result of Lemma 4.2.4 is the equation

$$
\langle u_i', u_j'\rangle \geq \frac{(\alpha\|x_i\| + 1)\big(\alpha\|x_j\| + 1\big)}{\|x_i\|\|x_j\|}p_1 + o_n(1),
$$

and hence

$$
\langle u_i', u_j'\rangle \geq p_1(1+\delta)^2 + o_n(1) = p_1 + \varepsilon + o_n(1),
$$

where $\varepsilon = p_1(2\delta + \delta^2)$. Since this was for any $i \neq j$, it follows that

$$
\liminf_{n\to\infty} \min_{i\neq j} \langle u_i', u_j'\rangle \geq p_1 + \varepsilon.
$$

For the case where $\gamma > \frac{1}{1-p_2}$, we formulate the result of Lemma 4.2.4 as

$$\langle u_i', u_j' \rangle = \frac{(\alpha\|x_i\| + 1)(\alpha\|x_j\| + 1)}{\|x_i\|\|x_j\|} \langle u_i, u_j \rangle + o_n(1)$$

and substitute the result of Lemma 4.2.3 to immediately obtain

$$\langle u_i', u_j' \rangle = \langle u_i, u_j \rangle + o_n(1).$$

Hence,

$$\lim_{n\to\infty} \langle u_i', u_j' \rangle = \langle u_i, u_j \rangle.$$

□

Corollary 4.2.1 (Phase Transition, Simplex Case): Assume there exist constants $q > 0$, $p \in (0,1)$ such that $\|x_i\|^2 = q$ and $\langle u_i, u_j \rangle = p$ for any $i \neq j$. Then,

$$\lim_{n\to\infty} \langle u_i', u_j' \rangle = \begin{cases} \frac{p(\alpha\sqrt{q}+1)^2}{\alpha^2 q + 2\alpha\sqrt{q}p + p} & \gamma < \frac{1}{1-p}, \\ \frac{p(\alpha\sqrt{q}+1)^2}{\alpha^2 q + \alpha\sqrt{q}(1+p) + \frac{1+3p}{4}} & \gamma = \frac{1}{1-p}, \\ p & \gamma > \frac{1}{1-p}. \end{cases}$$

Proof. This corresponds to the special case of Theorem 4.2.1 where $p_1 = p_2$.

If $\gamma < \frac{1}{1-p}$, the steps of the proof of Lemma 4.2.3 follow with equality, giving

$$\|x_i'\|^2 = \alpha^2 q + 2\alpha\sqrt{q}p + p + o_n(1).$$

Similar reasoning can be used with Lemma 4.2.4 to obtain

$$\langle x_i', x_j' \rangle = p(\alpha\sqrt{q} + 1)^2 + o_n(1).$$

Hence,

$$\lim_{n\to\infty} \langle u_i', u_j' \rangle = \frac{p(\alpha\sqrt{q}+1)^2}{\alpha^2 q + 2\alpha\sqrt{q}p + p}.$$

If $\gamma = \frac{1}{1-p}$, we first observe

$$\frac{\sum_{j\neq i} e^{s_{ij}}}{e^{\beta}} = \frac{(n-1)n^{\gamma p}}{n^{\gamma}} = n^{\gamma(p-1)+1} - n^{\gamma(p-1)} = 1 + o_n(1),$$

thus $Z_i = (2 + o_n(1))e^{\beta}$. Similar calculations as in Lemma 4.2.3 yield

$$\langle u_i, \mathrm{ATT}(u_i) \rangle = \frac{1+p}{2} + o_n(1),$$
$$\langle \mathrm{ATT}(u_i), \mathrm{ATT}(u_i) \rangle = \frac{1+3p}{4} + o_n(1).$$

Similar calculations as in Lemma 4.2.4 give

$$\langle u_i, \mathrm{ATT}(u_j)\rangle = p + o_n(1),$$
$$\langle \mathrm{ATT}(u_i), \mathrm{ATT}(u_j)\rangle = p + o_n(1).$$

By substituting these results into the inner product expansions, we obtain

$$\langle u_i', u_j'\rangle = \frac{\langle x_i', x_j'\rangle}{\|x_i'\|\|x_j'\|} = \frac{p(\alpha\sqrt{q}+1)^2}{\alpha^2 q + \alpha\sqrt{q}(1+p) + \frac{1+3p}{4}} + o_n(1).$$

If $\gamma > \frac{1}{1-p}$, Theorem 4.2.1 directly gives

$$\lim_{n\to\infty} \langle u_i, u_j\rangle = \langle u_i, u_j\rangle = p.$$

□

Since $p < 1$, the trailing term in the denominators of the $\gamma \geq \frac{1}{1-p}$ cases show that we have $\lim_{n\to\infty}\langle u_i', u_j'\rangle > p$. Thus, even when scaled appropriately, attention is still inherently a contractive operator.

Remark 4.2.1: In the absence of residual connections, i.e. $\alpha = 0$, we have

$$\lim_{n\to\infty} \langle u_i', u_j'\rangle = \begin{cases} 1 & \gamma < \frac{1}{1-p}, \\ \frac{4p}{1+3p} & \gamma = \frac{1}{1-p}, \\ p & \gamma > \frac{1}{1-p}. \end{cases}$$

The phase transition is especially clear in this case. In the subcritical regime, all token embeddings collapse onto the same direction, which implies that attention is behaving like an averaging operator, $\mathrm{ATT}(u_i) = \frac{1}{n}\sum_j u_j$. In the supercritical regime, the geometry of the embeddings is asymptotically unchanged and attention behaves like the identity map. In some sense, this provides a reason for why residual connections are crucial in avoiding this rank collapse phenomenon.

4.3 Backwards Pass

Theorem 4.3.1 (Phase Transition in Attention Gradient): Assume there exist constants $q_1, q_2 > 0$ and $p_1, p_2 \in (0,1)$ such that $q_1 \leq \|x_i\|^2 \leq q_2$ and $p_1 \leq \langle u_i, u_j\rangle \leq p_2$ for any $i \neq j$, with $p_1 = \langle u_i, u_j\rangle$ for some i, j. Let $\beta = \gamma \log n$ with $\gamma > 0$.

1. If $\gamma < \frac{1}{1-p_1}$,

$$\frac{1}{nd}\|\nabla_X X'\|^2 \leq \frac{4\gamma^2(\log n)^2}{q_1 d} + o_n(1).$$

2. If $\gamma > \frac{1}{1-p_2}$,

$$\frac{1}{nd}\|\nabla_X X'\|^2 \leq \frac{1}{q_2}\left(1 - \frac{1}{d}\right) + o_n(1).$$

Lemma 4.3.1: For any $i, k \in \{1, ..., n\}$ and $u, w \in \{1, ..., d\}$,

$$\frac{\partial(N(x_k))_w}{\partial(x_i)_u} = \delta_{ik} \frac{\delta_{wu}\|x_k\|^2 - (x_k)_w (x_k)_u}{\|x_k\|^3}$$

Proof. By the quotient rule,

$$\frac{\partial(N(x_k))_w}{\partial(x_i)_u} = \frac{\partial\left((x_k)_w \cdot \|x_k\|^{-1}\right)}{\partial(x_i)_u} = \delta_{ik} \frac{\delta_{wu}\|x_k\| - (x_k)_w \cdot \frac{(x_k)_u}{\|x_k\|}}{\|x_k\|^2} = \delta_{ik} \frac{\delta_{wu}\|x_k\|^2 - (x_k)_w (x_k)_u}{\|x_k\|^3}.$$

□

Lemma 4.3.2: For any $k, j \in \{1, ..., n\}$ and $w, v \in \{1, ..., d\}$,

$$\begin{aligned}\frac{\partial\left(\mathrm{ATT}(u_j)\right)_v}{\partial(u_k)_w} = &\left[\left(\delta_{kj}\beta\left(\sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}(u_m)_w (u_m)_v\right) + e^{\beta\langle u_j, u_k\rangle}\left(\beta(u_j)_w (u_k)_v + \delta_{wv}\right)\right) \cdot Z_j\right.\\ &\left. - \left(\delta_{kj}\beta\left(\sum_{l=1}^{n} e^{\beta\langle u_j, u_l\rangle}(u_l)_w\right) + \beta e^{\beta\langle u_j, u_k\rangle}(u_j)_w\right) \cdot \left(\sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}(u_m)_v\right)\right] \cdot Z_j^{-2}\end{aligned}$$

Proof. Recall

$$\left(\mathrm{ATT}(u_j)\right)_v = \frac{\sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}(u_m)_v}{\sum_{l=1}^{n} e^{\beta\langle u_j, u_l\rangle}} := \frac{(y_j)_v}{Z_j}$$

The derivative of the numerator,

$$\begin{aligned}\frac{\partial(y_j)_v}{\partial(u_k)_w} &= \sum_{m=1}^{n}\left[\left(\frac{\partial}{\partial(u_k)_w} e^{\beta\langle u_j, u_m\rangle}\right)(u_m)_v + e^{\beta\langle u_j, u_m\rangle}\frac{\partial}{\partial(u_k)_w}(u_m)_v\right]\\ &= \sum_{m=1}^{n}\left[\beta\left(\delta_{kj}(u_m)_w + \delta_{km}(u_j)_w\right)e^{\beta\langle u_j, u_m\rangle}(u_m)_v + e^{\beta\langle u_j, u_m\rangle}(\delta_{km}\delta_{wv})\right]\\ &= \left(\beta\delta_{kj}\sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}(u_m)_w (u_m)_v\right) + e^{\beta\langle u_j, u_k\rangle}\left(\beta(u_j)_w (u_k)_v + \delta_{wv}\right).\end{aligned}$$

The derivative of the denominator,

$$\frac{\partial Z_j}{\partial(u_k)_w} = \sum_{l=1}^{n}\beta\left(\delta_{kj}(u_l)_w + \delta_{kl}(u_j)_w\right)e^{\beta\langle u_j, u_l\rangle} = \left(\delta_{kj}\beta\sum_{l=1}^{n} e^{\beta\langle u_j, u_l\rangle}(u_l)_w\right) + \beta e^{\beta\langle u_j, u_k\rangle}(u_j)_w.$$

Applying the quotient rule yields the desired result. □

Lemma 4.3.3: The Jacobian of the attention operator has the form

$$\left(\frac{\partial\Big(\big(\mathrm{ATT}(N(x_j))\big)_v\Big)}{\partial(x_i)_u}\right)_{u,v=1}^{d} = \|x_i\|^{-1/2}\left[(\mathbf{R}_1+\mathbf{R}_2)Z_j - (\mathbf{U_1}+\mathbf{U_2})\otimes\mathbf{V}_j\right]\cdot Z_j^{-2},$$

where

$$\begin{aligned}
&\mathbf{R}_1 = \delta_{ij}\big(\mathbf{W}_j - u_i\otimes(\mathbf{W}_j u_j)\big), && \mathbf{R}_2 = e^{\beta\langle u_j,u_i\rangle}\Big(\big(-u_j+\beta\mathbf{P}_{u_i}u_j\big)\otimes u_i + I_d\Big),\\
&\mathbf{U}_1 = \delta_{ij}\beta\Big(\mathbf{P}_{u_i}\mathbf{V}_j\Big), && \mathbf{U}_2 = \beta e^{\beta\langle u_j,u_i\rangle}\Big(\mathbf{P}_{u_i}u_j\Big),\\
&\mathbf{V}_j = \sum_{m=1}^{n} e^{\beta\langle u_j,u_m\rangle}u_m, && \mathbf{W}_j = \sum_{m=1}^{n} e^{\beta\langle u_j,u_m\rangle}u_m\otimes u_m,\\
&\mathbf{P}_x y = y - \langle y,x\rangle x.
\end{aligned}$$

Proof. Let

$$J_{wv} = \frac{\partial\big(\mathrm{ATT}(N(x_j))\big)_v}{\partial(N(x_k))_w} = \frac{\partial\big(\mathrm{ATT}(u_j)\big)_v}{\partial(u_k)_w},$$

which was computed in the previous lemma. By the multivariate chain rule, we have

$$\begin{aligned}
\frac{\partial\Big(\big(\mathrm{ATT}(N(x_j))\big)_v\Big)}{\partial(x_i)_u} &= \sum_{k=1}^{n}\sum_{w=1}^{d} J_{wv}\cdot\frac{\partial\Big(N(x_k)_w\Big)}{\partial(x_i)_u}\\
&= \sum_{w=1}^{d} J_{wv}\cdot\frac{\delta_{wu}\|x_i\|^2 - (x_i)_w(x_i)_u}{\|x_i\|^3}\\
&= \|x_i\|^{-1}\left(J_{uv} - (u_i)_u\sum_{w=1}^{d} J_{wv}(u_i)_w\right).
\end{aligned}$$

We compute

$$\begin{aligned}
&\sum_{w=1}^{d} J_{wv}(u_i)_w\\
&= \left[\left(\delta_{ij}\beta\left(\sum_{m=1}^{n} e^{\beta\langle u_j,u_m\rangle}\langle u_m,u_i\rangle(u_m)_v\right) + e^{\beta\langle u_j,u_i\rangle}\Big((\beta\langle u_j,u_i\rangle+1)(u_i)_v\Big)\cdot Z_j\right.\\
&- \left.\left(\delta_{ij}\beta\left(\sum_{l=1}^{n} e^{\beta\langle u_j,u_l\rangle}\langle u_l,u_i\rangle\right) + \beta e^{\beta\langle u_j,u_i\rangle}\langle u_j,u_i\rangle\right)\cdot\left(\sum_{m=1}^{n} e^{\beta\langle u_j,u_m\rangle}(u_m)_v\right)\right]\cdot Z_j^{-2},
\end{aligned}$$

which then gives

$$\frac{\partial\Big(\big(\mathrm{ATT}(N(x_j))\big)_v\Big)}{\partial(x_i)_u} = \Bigg[\Bigg(\delta_{ij}\beta\Bigg(\sum_{m=1}^{n} e^{\beta\langle u_j,u_m\rangle}\Big((u_m)_u(u_m)_v - \langle u_m,u_i\rangle(u_m)_v(u_i)_u\Big)\Bigg)$$
$$+ e^{\beta\langle u_j,u_k\rangle}\Big(\beta(u_j)_u(u_k)_v + \delta_{uv} - \big(\beta\langle u_j,u_i\rangle+1\big)(u_i)_v(u_i)_u\Big)\Bigg)\cdot Z_j$$
$$-\Bigg(\delta_{ij}\beta\Bigg(\sum_{l=1}^{n} e^{\beta\langle u_j,u_l\rangle}\Big((u_l)_u - \langle u_l,u_i\rangle(u_i)_u\Big)\Bigg)$$
$$+\beta e^{\beta\langle u_j,u_k\rangle}\Big((u_j)_u - \langle u_j,u_i\rangle(u_i)_u\Big)\Bigg)\cdot\Bigg(\sum_{m=1}^{n} e^{\beta\langle u_j,u_m\rangle}(u_m)_v\Bigg)\Bigg]$$
$$\cdot Z_j^{-2}\cdot\|x_i\|^{-1}.$$

When written in matrix form (i.e. the transposed Jacobian), this becomes

$$\Bigg[\Big[\delta_{ij}\beta\big(\mathbf{W}_j - u_i\otimes(W_ju_i)\big) + e^{\beta\langle u_j,u_k\rangle}\big(\beta u_j\otimes u_i + I_d - \big(\beta\langle u_j,u_i\rangle+1\big)\,u_i\otimes u_i\big)\Big]\cdot Z_j$$
$$-\Big[\delta_{ij}\beta\big(\mathbf{V}_j - \langle\mathbf{V}_j,u_i\rangle u_i\big) + \beta e^{\beta\langle u_j,u_i\rangle}\big(u_j - \langle u_j,u_i\rangle u_i\big)\big)\Big]\otimes\mathbf{V}_j\Bigg]\cdot Z_j^{-2}\cdot\|x_i\|^{-1}$$
$$=\Bigg[\Big(\mathbf{R}_1 + e^{\beta\langle u_j,u_i\rangle}\Big(\big(-u_i+\beta\mathbf{P}_{u_i}u_j\big)\otimes u_i + I_d\Big)\Big)\cdot Z_j$$
$$-\Big[\delta_{ij}\beta\Big(\mathbf{P}_{u_i}\mathbf{V}_j\Big) + \beta e^{\beta\langle u_j,u_i\rangle}\Big(\mathbf{P}_{u_i}u_j\Big)\Big]\otimes\mathbf{V}_j\Bigg]\cdot Z_j^{-2}\cdot\|x_i\|^{-1}$$
$$=\Bigg[(\mathbf{R}_1+\mathbf{R}_2)\cdot Z_j + (\mathbf{U}_1+\mathbf{U}_2)\otimes\mathbf{V}_j\Bigg]\cdot Z_j^{-2}\cdot\|x_i\|^{-1},$$

where the intermediate steps follow from basic properties of the tensor product. □

We now prove Theorem 3.3.1.

Proof. Throughout this proof, we use the fact that $\|x\otimes y\|_F = \|x\|_2\|y\|_2$, as well as $\|A\|_F^2 = \mathrm{tr}(A^TA)$. The subscripts will be dropped since we are exclusively using the Frobenius norm for matrices and 2-norm for vectors. First consider the case when $\gamma > \frac{1}{1-p_2}$.

1. When $i \neq j$, $\mathbf{R}_1 Z_j^{-1} = 0$. When $i = j$, we obtain (using the form of $\mathbf{R}_1$ before simplification in the previous lemma)

$$\begin{aligned}
\|\mathbf{R}_1\| Z_j^{-1} &= \left\| \beta \sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle} (u_m \otimes u_m - \langle u_m, u_i\rangle u_i \otimes u_m) \right\| \cdot Z_j^{-1} \\
&= \left\| \beta \sum_{m\neq j} e^{\beta\langle u_j, u_m\rangle} (u_m \otimes u_m - \langle u_m, u_i\rangle u_i \otimes u_m) \right\| \cdot Z_j^{-1} \\
&\leq \beta \left[\sum_{m\neq j} e^{\beta p_2} \left(\|u_m\|^2 + \|u_m\|^2 \|u_i\|^2 \right) \right] \cdot Z_j^{-1} \\
&= 2\beta(n-1)e^{\beta p_2} \cdot Z_j^{-1} \\
&\leq 2\gamma \log(n) \cdot n^{\gamma p_2 + 1 - \gamma}(1 + o_n(1))
\end{aligned}$$

where the last inequality follows from Lemma 4.2.1. Since $\gamma p_2 + 1 - \gamma < 0$ by assumption, this value goes to 0 as $n \to \infty$.

2. When $i = j$, we notice that $\mathbf{P}_{u_i} u_j = 0$, hence

$$\mathbf{R}_2 Z_j^{-1} = e^{\beta} Z_j^{-1}(-u_i \otimes u_i + I_d) = (I_d - u_i \otimes u_i)(1 + o_n(1))$$

When $i \neq j$, naively bounding gives

$$\begin{aligned}
\|\mathbf{R}_2\| Z_j^{-1} &\leq e^{\beta p_2} Z_j^{-1} \left(\|u_i\|^2 + \beta \|\mathbf{P}_{u_i} u_j\| \|u_i\| + \|I_d\| \right) \\
&\leq n^{\gamma(p_2 - 1)} \left(1 + 2\gamma \log n + \sqrt{d} \right)(1 + o_n(1)),
\end{aligned}$$

which goes to 0 as $n \to \infty$ since $\gamma > 0$ and $p_2 - 1 < 0$.

3. When $i = j$, we have that $\mathbf{U}_2 = 0$ since $\mathbf{P}_{u_i} u_i = 0$, so

$$\begin{aligned}
&\left\| (\mathbf{U}_1 + \mathbf{U}_2) \otimes \mathbf{V}_j \right\| \cdot Z_j^{-2} = Z_j^{-2} \beta \left\| \mathbf{P}_{u_i} \mathbf{V}_j \right\| \left\| \mathbf{V}_j \right\| \\
&\leq Z_j^{-2} \beta \left(\sum_{m\neq j} e^{\beta\langle u_j, u_m\rangle} \left\| \mathbf{P}_{u_i} u_m \right\| \right) \cdot \left(e^{\beta} + \sum_{m\neq j} e^{\beta\langle u_j, u_m\rangle} \|u_m\| \right) \\
&\leq 2e^{-2\beta} \beta \left(n e^{\beta p_2} \right) \left(e^{\beta} + n e^{\beta p_2} \right)(1 + o_n(1)) \\
&\leq 2\gamma \log n \cdot n^{\gamma(p_2 - 1) + 1} \cdot \left(1 + n^{\gamma(p_2 - 1) + 1} \right)(1 + o_n(1)),
\end{aligned}$$

which goes to 0 as $n \to \infty$ since $\gamma(p_2 - 1) + 1 < 0$. When $i \neq j$, $\mathbf{U_1} = 0$, thus

$$\begin{aligned}
&\left\| (\mathbf{U}_1 + \mathbf{U}_2) \otimes \mathbf{V}_j \right\| \cdot Z_j^{-2} = Z_j^{-2} \beta e^{\beta\langle u_j, u_i\rangle} \left\| \mathbf{P}_{u_i} u_j \right\| \left\| \mathbf{V}_j \right\| \\
&\leq 2 Z_j^{-2} \beta e^{\beta p_2} \sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle} \|u_m\| \\
&\leq 2\beta e^{-2\beta} e^{\beta p_2} n e^{\beta}(1 + o_n(1)) \\
&\leq 2\gamma \log(n) n^{\gamma(p_2 - 1) + 1}(1 + o_n(1)),
\end{aligned}$$

which goes to 0 as $n \to \infty$ since $\gamma(p_2 - 1) + 1 < 0$. Combining all cases for $\gamma > \frac{1}{1 - p_2}$, we have that

$$\left(\frac{\partial\Big(\big(\mathrm{ATT}\big(N\big(x_j\big)\big)\big)_v\Big)}{\partial\big(x_i\big)_u}\right)_{u,v=1}^{d} \leq \frac{\delta_{ij}}{\|x_i\|}(I_d - u_i \otimes u_i)(1 + o_n(1))$$

$$= \frac{\delta_{ij}}{\|x_i\|}(I_d - u_i \otimes u_i) + \mathbf{o}_n(1) + o_n(1)I_d.$$

Noticing the block-diagonal structure of $\nabla_X X'$ and that $(u_i \otimes u_i)^2 = u_i \otimes u_i$,

$$\begin{aligned}
\frac{1}{nd}\|\nabla_X X'\|^2 &= \frac{1}{nd}\left(\sum_{i=1}^{n} \frac{1}{\|x_i\|}\|I_d - u_i \otimes u_i\|^2\right) + o_n(1) \\
&\geq \frac{1}{ndq_2}\left(\sum_{i=1}^{n} \mathrm{tr}(I_d - u_i \otimes u_i)\right) + o_n(1) \\
&= \frac{1}{ndq_2}\left(\sum_{i=1}^{n} d - \mathrm{tr}(u_i u_i^T)\right) + o_n(1) \\
&= \frac{1}{q_2}\left(1 - \frac{1}{d}\right) + o_n(1).
\end{aligned}$$

Next, consider the case when $\gamma < \frac{1}{1-p_2}$.

1. Once again, if $i \neq j$, $\mathbf{R_1} Z_j^{-1} = 0$. When $i = j$, we first note that $\left\|\mathbf{P}_{u_i} u_m\right\| \leq 1$, since $\mathbf{P}_{u_i} u_m$ gives the component of u_m orthogonal to u_i. Thus,

$$\begin{aligned}
\mathbf{R}_1 Z_j^{-1} &= \left\|\beta \sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}(u_m \otimes u_m - \langle u_m, u_i\rangle u_i \otimes u_m)\right\| \cdot Z_j^{-1} \\
&= \beta\left\|\sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}\left(\mathbf{P}_{u_i} u_m\right) \otimes u_m\right\| \cdot Z_j^{-1} \\
&\leq \beta Z_j^{-1} \sum_{m=1}^{n} e^{\beta\langle u_j, u_m\rangle}\left\|\mathbf{P}_{u_i} u_m\right\|\|u_m\| \\
&\leq \beta
\end{aligned}$$

2. As computed earlier, $\|I_d - u_i \otimes u_i\| = \sqrt{d-1}$, giving

$$\begin{aligned}
\|\mathbf{R}_2\| Z_j^{-1} &\leq Z_j^{-1} e^{\beta\langle u_j, u_i\rangle}\left\|\left(-u_i + \beta\mathbf{P}_{u_i} u_j\right) \otimes u_i + I_d\right\| \\
&\leq Z_j^{-1} e^{\beta\langle u_j, u_i\rangle}\left(\|I_d - u_i \otimes u_i\| + \beta\left\|\mathbf{P}_{u_i} u_j\right\|\|u_i\|\right) \\
&\leq Z_j^{-1} e^{\beta\langle u_j, u_i\rangle}\left(\sqrt{d-1} + \beta\right).
\end{aligned}$$

3. Note that $\left\|\mathbf{V}_j\right\| \leq Z_j$, and $\left\|\mathbf{P}_{u_i}\mathbf{V}_j\right\| \leq \left\|\mathbf{V}_j\right\| = Z_j$.

$$\begin{aligned}
\left\|(\mathbf{U}_1 + \mathbf{U}_2) \otimes \mathbf{V}_j\right\| \cdot Z_j^{-2} &\leq (\|\mathbf{U}_1\| + \|\mathbf{U}_2\|) \cdot Z_j^{-1} \\
&\leq Z_j^{-1}\delta_{ij}\beta\left\|\mathbf{P}_{u_i}\mathbf{V}_j\right\| + \beta e^{\beta\langle u_j, u_i\rangle}\left\|\mathbf{P}_{u_i} u_j\right\| \cdot Z_j^{-1} \\
&\leq \beta\left(\delta_{ij} + e^{\beta\langle u_j, u_i\rangle} Z_j^{-1}\right).
\end{aligned}$$

All together, this gives

$$
\begin{aligned}
&\left\| \left(\frac{\partial\Big(\big(\mathrm{ATT}(N(x_j))\big)_v\Big)}{\partial (x_i)_u} \right)_{u,v=1}^{d} \right\| \\
&\leq \frac{1}{\|x_i\|}\Big(\delta_{ij}\beta + Z_j^{-1} e^{\beta\langle u_j, u_i\rangle}\Big(\sqrt{d-1}+\beta\Big) + \beta\Big(\delta_{ij} + e^{\beta\langle u_j, u_i\rangle} Z_j^{-1}\Big)\Big) \\
&= \frac{1}{\|x_i\|}\Big(2\beta\delta_{ij} + \Big(2\beta + \sqrt{d}\Big) e^{\beta\langle u_j, u_i\rangle} Z_j^{-1}\Big).
\end{aligned}
$$

Using the properties of Z_j as shown in Lemma 4.2.1,

$$
\begin{aligned}
&\frac{1}{nd}\|\nabla_X X'\|^2 \\
&\leq \frac{1}{nd}\sum_{i,j=1}^{n} \frac{1}{\|x_i\|}\left(4\beta^2\delta_{ij} + 4\beta\delta_{ij}\Big(2\beta+\sqrt{d}\Big)\frac{e^{\beta\langle u_j, u_i\rangle}}{Z_j} + \Big(2\beta+\sqrt{d}\Big)^2 \cdot \frac{e^{2\beta\langle u_j, u_i\rangle}}{Z_j^2}\right) \\
&\leq \frac{1}{ndq_1}\left(n\cdot 4\beta^2 + O\big((\log n)^2\big)\left(\sum_{j=1}^{n}\frac{e^{\beta}}{Z_j}\right) + O\big((\log n)^2\big)\left(\sum_{i,j=1}^{n}\frac{e^{2\beta\langle u_j, u_i\rangle}}{Z_j^2}\right)\right) \\
&\leq \frac{(1+o_n(1))}{ndq_1}\left(4n\beta^2 + O\big((\log n)^2\big)\frac{n^{\gamma+1}}{n^{\gamma p_1+1}} + O\big((\log n)^2\big)\cdot\sum_{j=1}^{n}\left(\frac{n^{\gamma}}{n^{\gamma p_1+1}}\cdot\sum_{i=1}^{n}\frac{e^{\beta\langle u_j, u_i\rangle}}{Z_j}\right)\right) \\
&= \frac{(1+o_n(1))}{dq_1}\left(4\beta^2 + O\big((\log n)^2\big)n^{\gamma(1-p_1)-1} + O\big((\log n)^2\big)n^{\gamma(1-p_1)-1}\right) \\
&= 4\frac{\gamma^2(\log n)^2}{dq_1} + o_n(1),
\end{aligned}
$$

since $\gamma(1-p_1)-1<0$.

$\square$

5 Statistical Physics

$\sigma_Q^2 = \sigma_K^2 = \frac{\sigma_s}{d}$

Lemma 5.0.1 (Stein's Lemma): Let $Z = (Z_1, ..., Z_n)^T \sim \mathcal{N}(0, V)$ and $g : \mathbb{R}^n \to \mathbb{R}$ such that $g \in C^1$ and $\mathbb{E}[\|g(Z)\|], \mathbb{E}[\|\nabla g(Z)\|] < \infty$. Then,

$$\mathbb{E}[Z_i\, g(Z)] = \sum_{j=1}^{n} \mathrm{Cov}(Z_i, Z_j) \cdot \mathbb{E}\left[\frac{\partial}{\partial Z_j} g(Z)\right]$$

Proof. The pdf of Z has the form $p(z) = C \exp\left[-\frac{1}{2} z^T V^{-1} z\right]$. Thus,

$$\begin{aligned} \nabla p(z) &= -V^{-1} z p(z) \\ \Rightarrow z p(z) &= -V \nabla p(z) \\ \Rightarrow z_i p(z) &= -\sum_{j=1}^{n} V_{ij} \frac{\partial}{\partial z_j} p(z). \end{aligned}$$

Substituting the above equation into the calculation of expectation,

$$\mathbb{E}[Z_i g(Z)] = \int_{\mathbb{R}^n} z_i g(z) p(z)\, dz = -\sum_{j=1}^{n} V_{ij} \int_{\mathbb{R}^n} g(z) \frac{\partial}{\partial z_j} p(z)\, dz.$$

We then apply integration by parts along the dimension of z_j, noting that $g(z)$ vanishes when $z_j \to \pm\infty$ by assumption of finite expectation. This gives

$$\mathbb{E}[Z_i g(Z)] = \sum_{j=1}^{n} V_{ij} \int_{\mathbb{R}^n} \frac{\partial}{\partial z_j} g(z) p(z)\, dz = \sum_{j=1}^{n} \mathrm{Cov}(Z_i, Z_j) \cdot \mathbb{E}\left[\frac{\partial}{\partial z_j} g(Z)\right].$$

□

Lemma 5.0.2 (Derivative of Scaled Softmax): Let $S \in \mathbb{R}^{n \times n}$. For any $i, j, k \in \{1, ..., n\}$,

$$\frac{\partial a_{ij}}{\partial s_{ik}} = \frac{\partial}{\partial s_{ik}} \left(\frac{e^{h s_{ij}}}{\sum_{l=1}^{n} e^{h s_{il}}} \right) = h a_{ij} (\delta_{jk} - a_{ik}).$$

Proof. The derivative of the numerator is

$$\frac{\partial}{\partial s_{ik}} e^{h s_{ij}} = h e^{h s_{ij}} \delta_{jk}.$$

The derivative of the denominator Z_i is

$$\frac{\partial}{\partial s_{ik}} Z_i = h e^{h s^{ik}}.$$

By quotient rule,

$$\frac{\partial}{\partial s_{ik}}\left(\frac{e^{hs_{ij}}}{\sum_{l=1}^{n} e^{hs_{il}}}\right) = \frac{he^{hs_{ij}}\delta_{jk}Z - e^{hs_{ij}}he^{hs^{ik}}}{Z^2} = ha_{ij}(\delta_{jk} - a_{ik}).$$

□

Lemma 5.0.3: For large d, we have

$$s_{ij} \sim N(0, \sigma_s^2 q^2),$$

with covariance structure

$$\mathrm{Cov}(s_{ij}, s_{pq}) = \sigma_s^2 q_{ip} q_{jq}.$$

Proof. By definition,

$$\begin{aligned} s_{ij} &= \frac{1}{\sqrt{d}} \sum_{k=1}^{d} (W_Q x_i)_k (W_k x_j)_k \\ &= \frac{1}{\sqrt{d}} \sum_{k=1}^{d} \sum_{a,b=1}^{d} (W_Q)_{ka} (x_i)_a (W_K)_{kb} (x_j)_b \\ &:= \frac{1}{\sqrt{d}} \sum_{k=1}^{d} D_k. \end{aligned}$$

First, $\mathbb{E}[D_k] = 0$ since $\mathbb{E}\left[(W_Q)_{ka}\right] = 0$. As for the variance,

$$\begin{aligned} \mathbb{E}[D_k^2] &= \mathbb{E}\left[\left(\sum_{a=1}^{d} (W_Q)_{ka} (x_i)_a\right)^2\right] \cdot \mathbb{E}\left[\left(\sum_{b=1}^{d} (W_K)_{kb} (x_j)_b\right)^2\right] \\ &= \left(\sum_{a=1}^{d} \mathbb{E}\left[(W_Q)_{ka}^2\right] (x_i)_a^2\right) \left(\sum_{b=1}^{d} \mathbb{E}\left[(W_K)_{ka}^2\right] (x_j)_b^2\right) \\ &= \frac{\sigma_s^2}{d^2} \|x_i\|^2 \|x_j\|^2 \end{aligned}$$

Since the x_k's are i.i.d. Gaussian in $\mathbb{R}^d$, $\|x_k\|^2 \xrightarrow{p} qd$, for some $q > 0$ by concentration of norms. By the Central Limit Theorem, $s_{ij} \xrightarrow{d} N(0, \sigma_s^2 q^2)$. A similar calculation for covariance yields

$$
\begin{aligned}
\operatorname{Cov}(s_{ij}, s_{pq}) &= \mathbb{E}[s_{ij}s_{pq}] \\
&= \frac{1}{d}\sum_{k,a,b,l,\alpha,\beta=1}^{d} (x_i)_a (x_j)_b (x_p)_\alpha (x_q)_\beta \mathbb{E}\Big[(W_Q)_{ka}(W_Q)_{l\alpha}\Big]\mathbb{E}\Big[(W_K)_{kb}(W_K)_{l\beta}\Big] \\
&= \frac{1}{d}\sum_{k,a,b=1}^{d} (x_i)_a (x_j)_b (x_p)_a (x_q)_b \mathbb{E}\Big[(W_Q)_{ka}^2\Big]\mathbb{E}\Big[(W_K)_{ka}^2\Big] \\
&= \frac{\sigma_s^2}{d^3}\sum_{k=1}^{d} \langle x_i, x_p\rangle \langle x_j, x_q\rangle \\
&= \sigma_s^2 q_{ip} q_{jq}.
\end{aligned}
$$

□

Lemma 5.0.4: Define the function

$$
\Phi_n(\gamma, h) = \mathbb{E}[\log Z_i(\gamma, h)], \qquad Z_i(\gamma, h) = \sum_{j=1}^{n} e^{h s_{ij}},
$$

noting that Φ is a function of γ since the covariances of scores depend on γ. Then,

$$
\lim_{n\to\infty} \mathbb{E}\Big[Y_i^{(2)}\Big] = 1 - \frac{1}{\sigma_s^2 q(q-p)} \lim_{h\to 1}\lim_{n\to\infty} \frac{\partial}{\partial h}\Phi_n(\gamma, h).
$$

Proof. We first note that we can safely swap derivative and expectation in the case of

$$
\frac{\partial}{\partial h}\Phi_n(\gamma, h) = \mathbb{E}\left[\frac{\partial}{\partial h}\log Z_i(\gamma, h)\right] = \mathbb{E}\left[\frac{\sum_{j=1}^{n} s_{ij} e^{h s_{ij}}}{\sum_{k=1}^{n} e^{h s_{ik}}}\right] = \sum_{j=1}^{n} \mathbb{E}[s_{ij} a_{ij}].
$$

Since this is a sum of elements of the form $\mathbb{E}[s_{ij} g(s_i)]$ where s_i is Gaussian (by Lemma 4.3), we can apply Stein's Lemma to obtain

$$
\begin{aligned}
\frac{\partial}{\partial h}\Phi_n(\gamma, h) &= \sum_{j=1}^{n}\sum_{k=1}^{n} \operatorname{Cov}(s_{ij}, s_{ik})\mathbb{E}\left[\frac{\partial a_{ij}}{\partial s_{ik}}\right] \\
&= \sum_{j=1}^{n}\sum_{k=1}^{n} \sigma_s^2 q_{ii} q_{jk} \mathbb{E}[h a_{ij}(\delta_{jk} - a_{ik})].
\end{aligned}
$$

Justification needed Define $Y_{i,h}^{(2)} = \sum_{j=1}^{n} a_{ij}^2$.

$$\begin{aligned}
\frac{\partial}{\partial v}\Phi_i(\gamma,h) &\approx h\sigma_s^2 q\sum_{j=1}^{n}\sum_{k=1}^{n}\big(q\delta_{jk}+p(1-\delta_{jk})\big)\mathbb{E}\big[a_{ij}\delta_{jk}-a_{ij}a_{ik}\big] \\
&= h\sigma_s^2 q\sum_{j=1}^{n}\sum_{k=1}^{n}\mathbb{E}\big[qa_{ij}\delta_{jk}-qa_{ij}a_{ik}\delta_{jk}-pa_{ij}a_{ik}(1-\delta_{jk})\big] \\
&= h\sigma_s^2 q\sum_{j=1}^{n}\mathbb{E}\left[qa_{ij}-qa_{ij}^2-pa_{ij}\sum_{j'\neq j}a_{ij'}\right] \\
&= h\sigma_s^2 q\left(q\left(\mathbb{E}\left[\sum_{j=1}^{n}a_{ij}\right]-\mathbb{E}\big[a_{ij}^2\big]\right)-p\,\mathbb{E}\left[\sum_{j=1}^{n}a_{ij}(1-a_{ij})\right]\right) \\
&= h\sigma_s^2 q(q-p)\Big(1-\mathbb{E}\Big[Y_{i,h}^{(2)}\Big]\Big).
\end{aligned}$$

Rearranging and taking limits gives

$$\lim_{n\to\infty}\mathbb{E}\Big[Y_i^{(2)}\Big]=1-\frac{1}{\sigma_s^2 q(q-p)}\lim_{h\to 1}\lim_{n\to\infty}\frac{\partial}{\partial h}\Phi_n(\gamma,h).$$

□

Proof. We now use the replica method from statistical physics. ZN JSUTIFICATION

$$\mathbb{E}[Z_i^t(\gamma,h)]=\mathbb{E}\left[\left(\sum_{k=1}^{n}e^{hs_{ik}}\right)^t\right]=\sum_{k_1,\ldots,k_t=1}^{n}\mathbb{E}\left[\exp\left(h\sum_{a=1}^{t}s_{ik_a}\right)\right]$$

MGF justification

$$\begin{aligned}
\mathbb{E}[Z_i^t(\gamma,h)] &= \sum_{k_1,\ldots,k_t=1}^{n}\exp\left(\frac{h^2}{2}\sum_{a,b=1}^{t}\mathbb{E}\Big[s_{ik_a}s_{ik_b}\Big]\right) \\
&= \sum_{k_1,\ldots,k_t=1}^{n}\exp\left(\frac{h^2\sigma_s^2}{2}\sum_{a,b=1}^{t}q^2\delta_{k_ak_b}+qp\big(1-\delta_{k_ak_b}\big)\right) \\
&= \sum_{k_1,\ldots,k_t=1}^{n}\exp\left(\frac{h^2\sigma_s^2}{2}\sum_{a,b=1}^{t}\big(q(q-p)\delta_{k_ak_b}+qp\big)\right) \\
&= \sum_{k_1,\ldots,k_t=1}^{n}\exp\left(\frac{h^2\sigma_s^2}{2}q(q-p)\sum_{a,b=1}^{t}\delta_{k_ak_b}+O(t^2)\right)
\end{aligned}$$

Define the empirical overlap matrix Q with entries $Q_{ab}=\delta_{k_ak_b}$. Among all n^t replicas, there are

$$S(Q)=\sum_{k_1,\ldots,k_t=1}^{n}\prod_{a,b=1}^{t}\mathbb{1}\big(Q_{ab}=\delta_{k_ak_b}\big)$$

with a specific overlap matrix Q. Thus,

$$\mathbb{E}[Z_i^t(\gamma, h)] = \sum_Q S(Q) \exp\left(\frac{h^2\sigma_s^2}{2} q(q-p) \sum_{a,b=1}^{t} Q_{ab} + O(t^2)\right).$$

The large n behaviour of this expression is determined by the overlap matrix that best "balances" $S(Q)$ and $\sum_{a,b} Q_{ab}$. We let $n = e^N$ and control N.

Something about Q equally partitioned to give tx groups
- $S(Q)$ entropy gain from more independent clusters
- $\sum Q_{ab}$ energy gain from bigger clusters
- Replica system will be dominated by when these are balanced
- the only x that survives exponential weighting is the one where the net exponential growth rate is stationary

$$S(Q) \sim (\#\text{ tokens})^{\#\text{ groups}} = e^{Ntx}$$

$$\sum_{a,b=1}^{t} Q_{ab} = (\#\text{ group})(\#\text{ pairs per group}) = (tx) \cdot \left(\frac{1}{t}\right)^2 = \frac{x}{t}$$

$$\begin{aligned}\mathbb{E}[Z_i^t(\gamma, h)] &\sim \exp\left[\operatorname*{ext}_{0<x\leq 1}\left(Ntx + \frac{h^2\gamma\log(e^N)}{2} q(q-p)\frac{t}{x} + O(t^2)\right)\right] \\ &= \exp\left[Nt \operatorname*{ext}_{0<x\leq 1}\left(x + \frac{h^2\gamma}{2} q(q-p)\frac{1}{x} + O(t)\right)\right]\end{aligned}$$

Simplify

$$\frac{1}{Nt}\ln \mathbb{E}[Z_i^t(\gamma, h)]) = \operatorname*{ext}_{0<x\leq 1}\left(x + \frac{h^2\gamma}{2} q(q-p)\frac{1}{x}\right)$$

Extremum at

$$x_* = h\sqrt{\frac{\gamma q(q-p)}{2}}$$

$$\gamma_c =$$

$\square$

6 Unified Framework

6.1 Deterministic Scores

We now concern ourselves with a fixed score vector $s \in \mathbb{R}^n$. The following analysis is independent of the attention mechanism, in the sense that the scores are taken as given, where the specific token vectors and weight matrices used to obtain these scores are not of concern. Let s^* be the maximum of such scores, and define the *gaps* $\Delta_j = s^* - s_j$ from each score to the maximum. We refer to scores as *competitors*, as they are "competing" for the attention weight that is to be distributed by softmax. Thus, the attention weight of any given competitor can be rewritten as

$$a_j(\beta) = \frac{e^{-\beta \Delta_j}}{\sum_{k=1}^n e^{-\beta \Delta_k}},$$

which is equivalent to scaled softmax applied to the new score vector Δ with maximum 0. We define $Z(\beta) = \sum_{j=1}^n \exp(-\beta \Delta_j)$ as in previous sections, while noting that it is defined on gap scores and not raw scores in this context.

Definition 6.1.1 (CGF): The *cumulative gap-counting function* (CGF) is the number of competitors with gap at most t from the maximum,

$$N(t) = \sum_{j=1}^n \mathbb{1}(\Delta_j \leq t).$$

Now, $Z(\beta)$ can be rewritten as the Laplace transform of the CGF, since

$$Z(\beta) = \sum_{j=1}^n e^{-\beta \Delta_j} = \sum_{j=1}^n \beta \int_0^\infty \mathbb{1}(\Delta_j \leq t) e^{-\beta t}\, dt = \beta \int_0^\infty e^{-\beta t} N(t)\, dt.$$

This is clearly well-defined on finite n since $N(t) \leq n$. However, we immediately observe that if $N(t)$ is not of exponential order, the attention weights tend toward uniformity.

Definition 6.1.2: The *upper tail accumulation scale* Λ is the smallest exponent for which $N(t) \leq \exp(\Lambda t)$ for all $t > 0$. Equivalently,

$$\Lambda = \sup_{t>0} \frac{\log N(t)}{t}.$$

When $\Lambda < \infty$, a gap Δ_j is called a *contact gap* if $N(\Delta_j) = e^{\Lambda \Delta_j}$. Let Δ_Λ be the largest of all such contact gaps and define the *contact accumulation exponent*

$$\alpha = \frac{\log N(\Delta_\Lambda)}{\log n} \in [0, 1],$$

which satisfies $N(\Delta_\Lambda) = n^\alpha$.

Lemma 6.1.1: The Shannon entropy $H(\beta) = -\sum_{j=1}^n a_j(\beta) \log(a_j(\beta))$ satisfies

$$H(\beta) = \log Z(\beta) - \beta \frac{Z'(\beta)}{Z(\beta)}$$

Proof. Recall that $\sum_{j=1}^n a_j(\beta) = 1$. Thus,

$$H(\beta) = -\sum_{j=1}^n a_j(\beta)\left(-\beta \Delta_j - \log Z(\beta)\right) = \log Z(\beta) + \sum_{j=1}^n \beta \Delta_j \frac{e^{-\beta \Delta_j}}{Z(\beta)} = \log Z(\beta) - \beta \frac{Z'(\beta)}{Z(\beta)}.$$

□

Theorem 6.1.1: For each $n \geq 2$, let $s^{(n)} \in \mathbb{R}^n$ be a score vector of length n with corresponding upper tail accumulation scale Λ_n. For any positive sequence (β_n),

1. If $\beta_n / \Lambda_n \to 0$, then *top-two collapse* holds, i.e.

$$\lim_{n\to\infty} D_n(\beta_n) = 0,$$

where $D_n(\beta_n)$ is the difference between the largest and second largest attention weights $a^{(n)}(\beta_n)$.

2. If $\beta_n / \Lambda_n \to \infty$, then *entropy collapse* holds, i.e.

$$\lim_{n\to\infty} H_n(\beta_n) = 0,$$

where is the Shannon entropy of the attention weights $a^{(n)}(\beta_n)$.

Proof. Let $r_n = \beta_n / \Lambda_n$.

1. Let $\Delta_{\min}^{(n)}$ be the smallest non-zero gap, which necessarily has $\Delta_{\min}^{(n)} \leq \Delta_{\Lambda}^{(n)}$. The corresponding difference in attention weights can be bounded using $1 - e^{-x} \leq x$, giving

$$D_n(\beta_n) = \frac{1 - \exp\left(-\beta_n \Delta_{\min}^{(n)}\right)}{Z(\beta_n)} \leq \frac{\beta_n \Delta_{\min}^{(n)}}{Z(\beta_n)} \leq \frac{\beta_n \Delta_{\Lambda}^{(n)}}{Z(\beta_n)}.$$

By definition, we have that $\log\left(N\left(\Delta_{\Lambda}^{(n)}\right)\right) = \Lambda_n \Delta_{\Lambda}^{(n)} = \beta_n \Delta_{\Lambda}^{(n)} / r_n$. Substituting into and rearranging the definition of α_n gives $\beta_n \Delta_{\Lambda}^{(n)} = r_n \alpha_n \log n$. Since $e^{-\beta_n t}$ is monotone on the $N\left(\Delta_{\Lambda}^{(n)}\right)$ gaps with $\Delta_j \leq \Delta_{\Lambda}^{(n)}$, we have that

$$\begin{aligned} Z(\beta_n) &\geq N\left(\Delta_{\Lambda}^{(n)}\right) \exp\left(-\beta_n \Delta_{\Lambda}^{(n)}\right) \\ &= \exp\left(\beta_n \Delta_{\Lambda}^{(n)} / r_n\right) n^{r_n \alpha_n} \\ &= n^{\alpha_n (1 - r_n)}. \end{aligned}$$

Thus,

$$D_n(\beta_n) \leq \frac{\beta_n \Delta_{\Lambda}^{(n)}}{n^{\alpha_n(1-r_n)}} = r_n \alpha_n \log n \cdot n^{-\alpha_n(1-r_n)}.$$

Since $te^{-t} \leq e^{-1}$ for all $t \in \mathbb{R}$, setting $t = (1 - r_n) \cdot \alpha_n \log n$ yields

$$D_n(\beta_n) \leq \frac{r_n}{1 - r_n} \cdot e^{-1},$$

which goes to 0 as $n \to \infty$ since $r_n \to 0$ by assumption.

2. First observe that $Z(\beta_n) \geq 1$, since the gap between the maximum score and itself is always 0. Using the previously established integral form for $Z(\beta_n)$ as well as the fact that $N(t) \leq \exp(\Lambda^{(n)} t)$ by definition,

$$Z(\beta_n) = \beta_n \int_0^\infty e^{-\beta_n t} N(t)\, dt \le \beta_n \int_0^\infty e^{-(\beta_n - \Lambda_n)t}\, dt = \frac{\beta_n}{\beta_n - \Lambda_n} = \frac{r_n}{r_n - 1},$$

which goes to 1 as $n \to \infty$ since $r_n \to \infty$. Thus, $Z(\beta_n) \to 1$.

For any n, differentiating the integral form with respect to β gives

$$Z'(\beta) = \int_0^\infty e^{-\beta t} N(t)\, dt + \beta \frac{d}{d\beta}\left(\int_0^\infty e^{-\beta t} N(t)\, dt\right) = \beta^{-1} Z(\beta) - \beta \int_0^\infty t e^{-\beta t} N(t)\, dt,$$

which implies that

$$\begin{aligned} -\beta Z'(\beta) &= \beta^2 \int_0^\infty t e^{-\beta t} N(t)\, dt - Z(\beta) \\ &\le \beta^2 \int_0^\infty t e^{-(\beta - \Lambda_n)t}\, dt - Z(\beta) \\ &= \left(\frac{\beta}{\beta - \Lambda_n}\right)^2 - \left(\frac{\beta}{\beta - \Lambda_n}\right). \end{aligned}$$

Since $(\beta_n/(\beta_n - \Lambda_n)) \to 1$ and $-\beta Z'(\beta) = \beta \sum_{j=1}^n \Delta_j \exp(-\beta \Delta_j) \ge 0$, we have that $-\beta Z'(\beta) \to 0$. By Lemma 6.1.1,

$$\lim_{n\to\infty} H(\beta_n) = \lim_{n\to\infty} \left[\log Z(\beta_n) + \frac{1}{Z(\beta_n)}(-\beta(Z'(\beta)))\right] = 0.$$

$\square$

This establishes Λ_n as the order of the critical scaling for $s^{(n)}$. Denote $a_n \asymp b_n$ if there exists constants $c, C > 0$ such that for large enough n, $cb_n \le a_n \le Cb_n$.

Corollary 6.1.1 (Critical Scaling Exponent): Let $\xi_\Lambda > 0$. If $\Lambda_n \asymp (\log n)^\xi$, then ξ is unique and any non-collapsing scaling β_n must have $\beta_n \asymp (\log n)^{\xi_\Lambda}$.

Furthermore, if $\alpha_n \asymp (\log n)^{\xi_\alpha}$ and $\Delta_\Lambda^{(n)} \asymp (\log n)^{\xi_\Delta}$, then

$$\xi_\Lambda = \xi_\alpha - \xi_\Delta + 1.$$

Proof. If $\Lambda_n \asymp (\log n)^{\xi'}$ for $\xi' > 0$, then $1 \asymp (\log n)^{\xi - \xi'}$, which is only possible when $\xi = \xi'$. For a non-collapsing scaling, we must have $\beta_n/\Lambda_n \asymp 1$, which is only possible if $\beta_n \asymp (\log n)^\xi$.

Since $\Lambda_n = (\alpha_n \log n)/\Delta_\Lambda^{(n)}$,

$$\Lambda_n \asymp \frac{(\log n)^{\xi_\alpha} (\log n)}{(\log n)^{\xi_\Delta}} = (\log n)^{\xi_\alpha - \xi_\Delta + 1},$$

and by uniqueness of this exponent must mean that $\xi_\Lambda = \xi_\alpha - \xi_\Delta + 1$. $\square$

Consider the Simplex case from SECTION PREVIOUS. For any row i of the $n \times n$ similarity matrix,

$$a_{ij} = \begin{cases} q, & i = j, \\ p, & i \neq j. \end{cases}$$

The gap counting function is given by

$$N(t) = \begin{cases} 1, & t < q - p, \\ n, & t \geq q - p \end{cases}$$

which immediately gives $\Lambda_n = (q-p)^{-1} \log(n)$, $\alpha_n = 1$, $\Delta_\Lambda^{(n)} = q - p$. Thus, $\xi_\Lambda = 1$, $\xi_\alpha = 0$, and $\xi_\Delta = 0$, which satisfy the relation derived in Corollary 6.1.1. Furthermore, we see that the non-collapsing scaling has $\beta_n \asymp \log n$, which validates REFERENCE.

Corollary 6.1.2 (Relaxed Entropy Collapse): If we instead take

$$\Gamma = \limsup_{t>0} \frac{\log N(t)}{t}$$

Show attention realizability of scores